Nervous system disorders
Common (>1/100, <1/10): somnolence, headache
Uncommon (>1/1000, <1/100): dizziness, syncope

Cardiac disorders
Uncommon (>1/1000, <1/100): congestive heart failure, palpitations, bradycardia

Vascular disorders
Common (>1/100, <1/10): hypertension
Uncommon (>1/1000, <1/100): hypotension

Respiratory, thoracic and mediastinal disorders
Uncommon (>1/1000, <1/100): rhinitis, nasal dryness

Gastrointestinal disorders
Common (>1/100, <1/10): oral dryness
Uncommon (>1/1000, <1/100): taste perversion, diarrhoea, nausea

General disorders and administration site conditions
Common (>1/100, <1/10): asthenic conditions

Investigations
Common (>1/100, <1/10): LFTs abnormal

Immune system disorders
Uncommon (>1/1000, <1/100): allergic contact dermatitis

Postmarketing Experience

The following adverse drug reactions have been reported since COMBIGAN® has been marketed. Because they are reported voluntarily from a population of unknown size, estimates of frequency cannot be made:

Eye disorders: Vision blurred

Skin and subcutaneous tissue disorders: Erythema facial

Additional Adverse Reactions

Additional adverse events that have been seen with one of the components and may potentially occur also with COMBIGAN®:

Brimonidine

Eye disorders: blurring, ocular allergic reaction, corneal staining, conjunctival discharge, conjunctival papillae, iritis, iridocyclitis (anterior uveitis), miosis

Immune system disorders: hypersensitivity, skin reaction (including erythema, face oedema, pruritus, rash), vasodilatation

Psychiatric disorders: insomnia

Cardiac disorders: arrhythmias, tachycardia

Respiratory, thoracic and mediastinal disorders: upper respiratory symptoms

Gastrointestinal disorders: gastrointestinal symptoms

General disorders and administration site conditions: fatigue/drowsiness, and systemic allergic reactions

In cases where brimonidine has been used as part of the medical treatment of congenital glaucoma, symptoms of brimonidine overdose such as loss of consciousness, hypotension, hypotonia, bradycardia, hypothermia, cyanosis and apnoea have been reported in neonates and infants (less than 2 years of age) receiving brimonidine. A high incidence of somnolence has been reported in children 2 years of age and above, especially those in the 2-7 age range and/or weighing < 20 Kg.

Timolol

Eye disorders: decreased corneal sensitivity, blepharoptosis, diplopia, keratitis, ptosis, choroidal detachment (following filtration surgery), refractive changes (due to withdrawal of miotic therapy in some cases), cystoid macular oedema, pseudopemphigoid

Psychiatric disorders: insomnia, nightmares, behavioral changes and psychic disturbances including anxiety, confusion, disorientation, hallucinations, memory loss, nervousness

Nervous system disorders: increase in signs and symptoms of myasthenia gravis, cerebral ischaemia, cerebral vascular accident, insomnia, nightmares, fatigue and paresthesia

Ear and labyrinth disorders: tinnitus

Cardiac disorders: heart block, cardiac arrest, arrhythmia, bradycardia, atrioventricular block, cardiac failure, chest pain, oedema, pulmonary oedema, worsening of angina pectoris

Vascular disorders: syncope

Respiratory, thoracic and mediastinal disorders: bronchospasm (predominantly in patients with pre-existing bronchospastic disease) dyspnoea, cough, respiratory failure, exacerbation of asthma, nasal congestion, upper respiratory infection

Gastrointestinal disorders: dyspepsia, abdominal pain, anorexia, vomiting

Skin and subcutaneous tissue disorders: alopecia, psoriasiform rash or exacerbation of psoriasis, skin rash

Musculoskeletal, connective tissue and bone disorders: myalgia

Renal and urinary disorders: Peyronie's disease, retroperitoneal fibrosis

Immune system disorders: Signs and symptoms of systemic allergic reactions including angioedema, generalized and localized rash, pruritus, urticaria; Systemic lupus erythematosus

Endocrine disorders: Masked symptoms of hypoglycaemia in diabetes patients

Other: Decreased libido, sexual dysfunction

OVERDOSAGE

There is limited data available of overdosage in humans with the use of COMBIGAN®. Bradycardia has been reported in association with use of a higher than recommended dose. If overdosage occurs, treatment should be symptomatic and supportive; a patent airway should be maintained.

There have been reports of inadvertent overdosage with timolol ophthalmic solution resulting in systemic effects similar to those seen with systemic beta-adrenergic blocking agents such as dizziness, headache, shortness of breath, bradycardia, hypotension, bronchospasm, and cardiac arrest. An *in vitro* haemodialysis study, using ^{14}C timolol added to human plasma or whole blood showed that timolol was readily dialysed from these fluids; however, a study of patients with renal failure showed that timolol did not dialyse readily.

Brimonidine

Ophthalmic overdose:

In cases where brimonidine has been used as part of the medical treatment of congenital glaucoma, symptoms of brimonidine overdose such as a loss of consciousness, hypotension, hypotonia, bradycardia, hypothermia, cyanosis and apnoea, coma, lethargy, pallor, respiratory depression, and somnolence have been reported in neonates and infants (less than 2 years of age) receiving brimonidine.

Systemic overdose resulting from accidental ingestion:

Several reports of serious adverse events following inadvertent ingestion of brimonidine by paediatric subjects have been published or reported to Allergan. The subjects experienced symptoms of CNS depression, typically temporary coma or low level of consciousness, hypotonia, bradycardia, hypothermia and apnoea and required admission to intensive care with intubation if indicated. All subjects were reported to have made a full recovery, usually within 6-24 hours.

Two cases of adverse effects following inadvertent ingestion of 9-10 drops of brimonidine by adult subjects have been received. The subjects experienced a hypotensive episode, followed in one instance by rebound hypertension approximately 8 hours after ingestion. Both subjects were reported to have made a full recovery within 24 hours. No adverse effects were noted in a third subject who ingested an unknown amount of brimonidine orally.

Oral overdoses of other alpha-2-agonists have been reported to cause symptoms such as hypotension, asthenia, vomiting, lethargy, sedation, bradycardia, arrhythmias, miosis, apnoea, hypotonia, hypothermia, respiratory depression and seizure.

Timolol

Symptoms of systemic timolol overdose are: bradycardia, hypotension, bronchospasm, headache, dizziness and cardiac arrest. A study of patients showed that timolol did not dialyse readily.

If overdose occurs treatment should be symptomatic and supportive.

DOSAGE AND ADMINISTRATION

Use in children and adolescents

COMBIGAN® is contraindicated in neonates and infants (less than 2 years of age).

The safety and effectiveness of COMBIGAN® in children and adolescents (2 to 17 years of age) have not been established and therefore, its use is not recommended in children or adolescents.

Recommended dosage in adults (including the elderly)

The recommended dose is one drop of COMBIGAN® in the affected eye(s) twice daily, approximately 12 hours apart. If more than one topical ophthalmic product is to be used, the different products should be instilled at least 5 minutes apart. As with any eye drops, to reduce possible systemic absorption, it is recommended that the lachrymal sac be compressed at the medial canthus (punctual occlusion) for one minute. This should be performed immediately following the instillation of each drop. To avoid contamination of the eye or drops do not allow the dropper tip to come into contact with any surface.

HOW SUPPLIED

COMBIGAN® eye drops are supplied sterile in white opaque plastic dropper bottles of 5 mL.

NOTE:

Store below 30°C. On prescription only. Keep out of reach of children. Keep the bottle in the outer carton. Discard unused contents 4 weeks after opening.

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Manufactured by:
Allergan Pharmaceuticals Ireland
Westport, Ireland

Date of revision: August 2023

20081046

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COMBIGAN® eye drops

brimonidine tartrate 2 mg/mL and timolol 5 mg/mL

DESCRIPTION AND COMPOSITION

Each mL contains: 2.0 mg brimonidine tartrate (equivalent to 1.3 mg of brimonidine) and 5.0 mg timolol (equivalent to 6.8 mg of timolol maleate) with benzalkonium chloride, sodium phosphate monobasic monohydrate, sodium phosphate dibasic heptahydrate, hydrochloric acid or sodium hydroxide to adjust pH, and purified water. COMBIGAN® is a clear, greenish-yellow to light greenish-yellow solution.

CLINICAL PHARMACOLOGY

Pharmacodynamic properties

Pharmacotherapeutic group:

Ophthalmological – antiglaucoma preparations and miotics -beta blocking agents - timolol, combinations

ATC code: SO1ED 51

Mechanism of action:

COMBIGAN® consists of two active substances: brimonidine tartrate and timolol maleate. These two components decrease elevated intraocular pressure (IOP) by complementary mechanisms of action and the combined effect results in additional IOP reduction compared to either compound administered alone. COMBIGAN® has a rapid onset of action.

Brimonidine tartrate is an alpha-2 adrenergic receptor agonist that is 1000-fold more selective for the alpha-2 adrenoreceptor than the alpha-1 adrenoreceptor. This selectivity results in no mydriasis and the absence of vasoconstriction in microvessels associated with human retinal xenografts.

It is thought that brimonidine tartrate lowers IOP by suppressing aqueous humour inflow and enhancing uveoscleral outflow. Timolol is a $beta_1$ and $beta_2$ non-selective adrenergic receptor blocking agent that does not have significant intrinsic sympathomimetic, direct myocardial depressant, or local anaesthetic (membrane-stabilising) activity. Timolol lowers IOP by reducing aqueous humour formation. The precise mechanism of action is not clearly established, but inhibition of the increased cyclic AMP synthesis caused by endogenous beta-adrenergic stimulation is probable.

Clinical effects:

In three controlled, double-masked clinical studies, COMBIGAN® (twice daily) produced a clinically meaningful additive decrease in mean diurnal IOP compared with timolol (twice daily) and brimonidine (twice daily or three times a day) when administered as monotherapy.

In a study in patients whose IOP was insufficiently controlled following a minimal 3-week run-in on any monotherapy, additional decreases in mean diurnal IOP of 4.5, 3.3 and 3.5 mmHg were observed during 3 months of treatment for COMBIGAN® (twice daily), timolol (twice daily) and brimonidine (twice daily), respectively. In this study, at trough, a significant additional decrease in IOP could only be demonstrated on comparison with brimonidine but not with timolol, however a positive trend was seen with superiority at all other timepoints. In the pooled data of the other two trials statistical superiority versus timolol was seen throughout.

In addition, the IOP-lowering effect of COMBIGAN® was consistently non-inferior to that achieved by adjunctive therapy of brimonidine and timolol (all twice daily).

The IOP-lowering effect of COMBIGAN® has been shown to be maintained in double-masked studies of up to 12 months.

Pharmacokinetic properties

COMBIGAN®:

Plasma brimonidine and timolol concentrations were determined in a crossover study comparing the monotherapy treatments to COMBIGAN® treatment in healthy subjects. There were no statistically significant differences in brimonidine or timolol AUC between COMBIGAN® and the respective monotherapy treatments. Mean plasma C_{max} values for brimonidine and timolol following dosing with COMBIGAN® were 0.0327 and 0.406 ng/mL respectively.

Colors:
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Brimonidine:

After ocular administration of 0.2% eye drops solution in humans, plasma brimonidine concentrations are low. Brimonidine is not extensively metabolised in the human eye and human plasma protein binding is approximately 29%. The mean apparent half-life in the systemic circulation was approximately 3 hours after topical dosing in man.

Following oral administration to man, brimonidine is well absorbed and rapidly eliminated. The major part of the dose (around 74% of the dose) was excreted as metabolites in urine within five days; no unchanged drug was detected in urine. In vitro studies, using animal and human liver, indicate that the metabolism is mediated largely by aldehyde oxidase and cytochrome P450. Hence, the systemic elimination seems to be primarily hepatic metabolism.

Brimonidine binds extensively and reversibly to melanin in ocular tissues without any untoward effects. Accumulation does not occur in the absence of melanin. Brimonidine is not metabolised to a great extent in human eyes.

Timolol:

After ocular administration of a 0.5% eye drops solution in humans undergoing cataract surgery, peak timolol concentration was 898 ng/mL in the aqueous humour at one hour post-dose. Part of the dose is absorbed systemically where it is extensively metabolised in the liver. The half-life of timolol in plasma is about 7 hours. Timolol is partially metabolised by the liver with timolol and its metabolites excreted by the kidney. Timolol is not extensively bound to plasma protein.

INDICATIONS AND USAGE

Reduction of intraocular pressure (IOP) in patients with chronic open-angle glaucoma or ocular hypertension who are insufficiently responsive to topical beta-blockers.

CONTRAINDICATIONS

- Reactive airway disease including bronchial asthma or a history of bronchial asthma, severe chronic obstructive pulmonary disease
- Sinus bradycardia, sick sinus syndrome, sino-atrial nodal block, second or third degree atrioventricular block, not controlled with a pace-maker, overt cardiac failure, cardiogenic shock
- Use in neonates and infants (less than 2 years of age); see Paediatric Use section.
- Patients receiving monoamine oxidase (MAO) inhibitor therapy
- Patients on antidepressants which affect noradrenergic transmission (e.g. tricyclic antidepressants and mianserin)
- Hypersensitivity to the active substances or any of the excipients

WARNINGS AND PRECAUTIONS

Children of 2 years of age and above, especially those in the 2-7 age range and/or weighing <20 Kg, should be treated with caution and closely monitored due to the high incidence of somnolence. The safety and effectiveness of COMBIGAN® in children and adolescents (2 to 7 years of age) have not been established.

Delayed ocular hypersensitivity reactions have been reported with brimonidine tartrate ophthalmic solution 0.2%, with some reported to be associated with an increase in IOP.

Some patients have experienced ocular allergic type reactions (allergic conjunctivitis and allergic blepharitis) with COMBIGAN® in clinical trials. Allergic conjunctivitis was seen in 5.2% of patients. Onset was typically between 3 and 9 months resulting in an overall discontinuation rate of 3.1%. Allergic blepharitis was uncommonly reported (<1%). If allergic reactions are observed, treatment with COMBIGAN® should be discontinued.

Like other topically applied ophthalmic agents, COMBIGAN® may be absorbed systemically. No enhancement of the systemic absorption of the individual active substances has been observed. Due to the beta-adrenergic component, timolol, the same types of cardiovascular and pulmonary adverse reactions as seen with systemic beta-blockers may occur. Incidence of systemic ADRs after topical ophthalmic administration is lower than for systemic administration. To reduce the systemic absorption, see section DOSAGE AND ADMINISTRATION.

Cardiac disorders:

In patients with cardiovascular diseases (e.g., coronary heart disease, Prinzmetal's angina and cardiac failure) and hypotension therapy with beta-blockers should be critically assessed with the therapy with other active substances should be considered.

Patients with cardiovascular diseases should be watched for signs of deterioration of these diseases and of adverse reactions.

Due to its negative effect on conduction time, beta-blockers should only be given with caution to patients with first degree heart block.

As with systemic beta-blockers, if discontinuation of treatment is needed in patients with coronary heart disease, therapy should be withdrawn gradually to avoid rhythm disorders, myocardial infarct or sudden death.

Respiratory disorders:

Respiratory reactions, including death due to bronchospasm in patients with asthma, have been reported following administration of some ophthalmic beta-blockers. COMBIGAN® should be used with caution in patients with mild/moderate chronic obstructive pulmonary disease (COPD) and only if the potential benefit outweighs the potential risk.

Surgical anaesthesia:

Beta-blocking ophthalmological preparations may block beta-agonist effects e.g. of adrenaline. The anaesthetist must be informed if the patient is using COMBIGAN®.

In patients with severe renal impairment on dialysis, treatment with timolol has been associated with pronounced hypotension.

Other beta-blocking agents:

The effect on intra-ocular pressure or the known effects of systemic beta-blockade may be potentiated when timolol is given to the patient already receiving a systemic beta-blocking agent. The response of these patients should be closely observed. The use of two topical beta-adrenergic blocking agents is not recommended (see section DRUG INTERACTIONS).

Hyperthyroidism:

Beta-blockers may also mask the signs of hyperthyroidism.

COMBIGAN® must be used with caution in patients with metabolic acidosis and untreated phaeochromocytoma.

Corneal diseases:

Ophthalmic beta-blockers may induce dryness of eyes. Patients with corneal diseases should be treated with caution.

Hypoglycemia/diabetes:

Beta-blockers should be administered with caution in patients subject to spontaneous hypoglycaemia or to uncontrolled diabetic patients (especially those with labile diabetes) as beta-blockers may mask the signs and symptoms of acute hypoglycaemia. The indicatory signs of acute hypoglycaemia may be masked, in particular tachycardia, palpitations and sweating.

Vascular disorders:

Patients with severe peripheral circulatory disturbance/disorders (i.e. Raynaud's phenomenon should be treated with caution.

Anaphylactic reactions:

While taking beta-blockers, patients with a history of atopy or a history of severe anaphylactic reaction to a variety of allergens may be more reactive to repeated challenge with such allergens. Such patients may be unresponsive to the usual dose of adrenaline used to treat anaphylactic reactions.

Choroidal detachment:

Choroidal detachment after filtration procedures has been reported with administration of aqueous suppressant therapy (e.g. timolol, acetazolamide).

The preservative in COMBIGAN®, benzalkonium chloride, may cause eye irritation. Patients wearing soft (hydrophilic) contact lenses should be instructed to remove contact lenses prior to application and wait at least 15 minutes after instilling COMBIGAN® before reinsertion. Benzalkonium chloride is known to discolour soft contact lenses. Avoid contact with soft contact lenses.

Patients should be instructed to avoid allowing the tip of the dispensing container to contact the eye or surrounding structures to avoid eye injury and contamination of eye drops.

COMBIGAN® has not been studied in patients with closed-angle glaucoma.

COMBIGAN® has not been studied in patients with hepatic or renal impairment. Therefore, caution should be used in treating such patients.

DRUG INTERACTIONS

No interaction studies have been performed with COMBIGAN®. Although specific drug interactions studies have not been conducted with COMBIGAN®, the theoretical possibility of an additive or potentiating effect with CNS depressants (alcohol, barbiturates, opiates, sedatives, or anaesthetics) should be considered.

There is potential for additive effects resulting in hypotension, and/or marked bradycardia when beta-blocker eye drops are administered concomitantly with oral calcium channel blockers, guanethidine, beta-blocking agents, anti-arrhythmics (including amiodarone), digitalis glycosides, parasympathomimetics, and other anti-hypertensives. After the application of brimonidine, very rare (<1 in 10,000) cases of hypotension have been reported. Caution is therefore advised when using COMBIGAN® with systemic antihypertensives. Although timolol has little or no effect on the size of the pupil, mydriasis has occasionally been reported when timolol has been used with mydriatic agents such as adrenaline. Beta-blockers may increase the hypoglycaemic effect of antidiabetic agents. Beta-blockers can mask the signs and symptoms of hypoglycaemia.

The hypertensive reaction to sudden withdrawal of clonidine can be potentiated when taking beta-blockers.

Potentiated systemic beta-blockade (e.g. decreased heart rate, depression) has been reported during combined treatment with CYP2D6 inhibitors [e.g., quinidine, selective serotonin reuptake inhibitors (SSRIs)] and timolol, possibly because quinidine inhibits the metabolism of timolol via the P450 enzyme, CYP2D6.

Concomitant use of a beta-blocker with anaesthetic drugs may attenuate compensatory tachycardia and increase the risk of hypotension, and therefore the anaesthetist must be informed if the patient is using COMBIGAN®. Caution must be exercised if COMBIGAN® is used concomitantly with iodine contrast products or intravenously administered lidocaine. Cimetidine, hydralazine and alcohol may increase the plasma concentrations of timolol.

The hypertensive reaction to sudden withdrawal of clonidine can be potentiated when taking beta-blockers. No data on the level of circulating catecholamines after COMBIGAN® administration are available. Caution, however, is advised in patients taking tricyclic antidepressants which can affect the metabolism and uptake of circulating amines e.g. chlorpromazine, methylphenidate, reserpine.

Caution is advised when initiating (or changing the dose of) a concomitant systemic agent (irrespective of pharmaceutical form) which may interact with α-adrenergic agonists or interfere with their activity i.e. agonists or antagonists of the adrenergic receptor (e.g. isoprenaline, prazosin).

Although specific drug interactions studies have not been conducted with COMBIGAN®, the theoretical possibility of an additive IOP lowering effect with prostamides, prostaglandins, carbonic anhydrase inhibitors and pilocarpine should be considered. Concomitant administration of MAO inhibitors is contraindicated. Patients who have been receiving MAOI therapy should wait 14 days after discontinuation before commencing treatment with COMBIGAN®.

Patients who are receiving a systemic (e.g., oral or intravenous) beta-adrenergic blocking agent and COMBIGAN® should be observed for potential additive effects of beta-blockade, both systemic and on intraocular pressure.

Preclinical safety data:

The ocular and systemic safety profile of the individual components is well established.

Pregnancy:

There are no adequate and well-controlled studies with COMBIGAN® in pregnant women. Because animal reproduction studies are not always predictive of human response, COMBIGAN® should be used during pregnancy only if the potential benefit to the mother justifies the potential risk to the foetus.

Brimonidine tartrate

No adequate clinical data on exposed pregnancies are available. Animal studies have shown reproductive toxicity at high maternotoxic doses.

Timolol

Epidemiological studies have not revealed malformative effects but shown a risk for intra uterine growth retardation when beta-blockers are administered by the oral route.

In addition, signs and symptoms of beta-blockade (e.g. bradycardia, hypotension, respiratory distress and hypoglycaemia) have been observed in the neonate when beta-blockers have been administered until delivery. If COMBIGAN® is administered until delivery, the neonate should be carefully monitored during the first days of life. Animal studies with timolol have shown reproductive toxicity at doses significantly higher than would be used in clinical practice. COMBIGAN® should not be used during pregnancy unless clearly necessary.

Lactation:

Timolol is excreted in human milk. It is not known if brimonidine is excreted in human milk but is excreted in the milk of the lactating rat. Therefore, COMBIGAN® should not be used by women breast- feeding infants.

Geriatric Use:

No overall differences in safety and effectiveness have been observed between elderly and other adult patients.

Effects on ability to drive and use machines:

COMBIGAN® has minor influence on the ability to drive and use machines.

COMBIGAN® may cause transient blurring of vision, visual disturbance, fatigue and/or drowsiness which may impair the ability to drive or operate machines. Patients who engage in activities such as driving and operating machinery should be cautioned of the potential for a decrease in mental alertness. The patient should wait until these symptoms have cleared before driving or using machinery.

ADVERSE EVENTS

Based on 12 month clinical data, the most commonly reported ADRs were conjunctival hyperaemia (approximately 15% of patients) and burning sensation in the eye (approximately 11% of patients). The majority of cases was mild and led to discontinuation rates of only 3.4% and 0.5% respectively.

The following adverse drug reactions were reported during clinical trials with COMBIGAN®:

Eye disorders

Very Common (>1/10): conjunctival hyperaemia, burning sensation

Common (>1/100, <1/10): stinging sensation in the eye, eye pruritus, allergic conjunctivitis, conjunctival folliculosis, visual disturbance, blepharitis, epiphora, corneal erosion, superficial punctate keratitis, erythema eyelid, eye dryness, eye discharge, eye pain, eye irritation, foreign body sensation, eyelid oedema, eyelid pruritus

Uncommon (>1/1000, <1/100): visual acuity worsened, conjunctival oedema, follicular conjunctivitis, allergic blepharitis, conjunctivitis, vitreous floater, asthenopia, photophobia, papillary hypertrophy, eyelid pain, conjunctival blanching, corneal oedema, corneal infiltrates, vitreous detachment

Psychiatric disorders

Common (>1/100, <1/10): depression

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